

A Fiber Phase-Coherence Certificate for Post-Bin Determinant Energy

Exact Mean-Fluctuation Splitting, Sector Lower Bounds, and the Remaining Arithmetic Gate

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Abstract

TPC-110 identified the post-bin determinant energy as the squared norm of the literal coefficient after summation on equal-determinant fibers. Its kernel-angle factorization is exact but does not say what coefficient-specific fact could force a positive angle. We give two such refinements. First, the pre-bin energy splits orthogonally into fiber-mean coherence and within-fiber fluctuation, while the determinant energy is exactly the fiber-size-weighted coherent part. Second, a phase-sector certificate lower-bounds a fiber sum when a good absolute-mass component lies in one half-plane and the remaining absolute mass is smaller than the projected good mass. The missing amplification is measured by the effective participation ratio of the good mass. At the growing scale, a diagonal mass exponent λ_E , participation gain γ_R , and phase-margin exponent λ_ϕ imply

$$\lambda_D^{\text{cert}} = (\lambda_E + 2\lambda_\phi - \gamma_R)_+.$$

Combining this with the TPC-112 compatibility cone gives an explicit three-input requirement. None of the growing inputs or the zero-mode saving is proved here. The result turns an opaque kernel angle into literal coefficient tests; it is an exact L0/L1 gate, not an L2 arithmetic advance.

1 Literal post-bin fibers

Fix the prescribed nonzero shift h_0 . Import the inverse-aggregated actual carrier $\mathcal{T}_X$, determinant label $\nu : \mathcal{T}_X \rightarrow \mathcal{N}_X$, and literal coefficients $u = (u_t)_{t \in \mathcal{T}_X}$ from TPC-110 [1]. For every nonempty fiber

$$F_n = \{t \in \mathcal{T}_X : \nu(t) = n\}, \quad m_n = |F_n|,$$

put

$$A_X(n) = \sum_{t \in F_n} u_t, \quad E_X = \sum_{t \in \mathcal{T}_X} |u_t|^2, \quad D_X = \sum_{n \in \mathcal{N}_X} |A_X(n)|^2.$$

All coefficients, determinant labels, and fibers are literal. No random phases, ambient support, or averaged shift is inserted.

For each fiber define

$$\mu_n = \frac{1}{m_n} \sum_{t \in F_n} u_t, \quad C_n = m_n |\mu_n|^2, \quad V_n = \sum_{t \in F_n} |u_t - \mu_n|^2.$$

Here C_n is the squared norm of the constant-on-fiber projection and V_n is its orthogonal complement.

Theorem 1.1 (Exact mean-fluctuation splitting). *One has*

$$\boxed{E_X = C_X + V_X}, \quad C_X = \sum_n C_n, \quad V_X = \sum_n V_n,$$

and

$$\boxed{D_X = \sum_n m_n C_n}.$$

If $C_X > 0$, the coherent effective multiplicity

$$\bar{m}_X^{\text{coh}} = \frac{\sum_n m_n C_n}{\sum_n C_n}$$

satisfies

$$m_{\min} \leq \bar{m}_X^{\text{coh}} \leq m_{\max}, \quad D_X = C_X \bar{m}_X^{\text{coh}}.$$

Proof. On F_n , the constant vector $(\mu_n)_{t \in F_n}$ is orthogonal to $(u_t - \mu_n)_{t \in F_n}$, because the latter sums to zero. Pythagoras gives

$$\sum_{t \in F_n} |u_t|^2 = m_n |\mu_n|^2 + V_n.$$

Also $A_X(n) = m_n \mu_n$, hence

$$|A_X(n)|^2 = m_n^2 |\mu_n|^2 = m_n C_n.$$

Summation proves both identities. The effective multiplicity is a weighted average of the nonempty m_n . $\square$

Remark 1.2. This is equivalent to the TPC-110 kernel-angle identity, but exposes the missing input locally: a determinant lower bound requires the literal coefficient to have substantial fiber means, not merely many collisions or large pre-bin energy.

2 A coefficient-specific phase certificate

The next lemma is deterministic. It is stated so that every term can be checked on an actual fiber.

Lemma 2.1 (Good-sector minus bad-mass bound). *Let $F = G \sqcup B$ be one fiber and let $u_t \in \mathbb{C}$. Suppose that there are a unit complex number ζ , a number $\kappa \in [0, 1]$, and $\theta \geq 0$ such that*

$$\Re(\bar{\zeta} u_t) \geq \kappa |u_t| \quad (t \in G),$$

and

$$\sum_{t \in B} |u_t| \leq \theta \sum_{t \in G} |u_t|.$$

Then

$$\boxed{\left| \sum_{t \in F} u_t \right| \geq (\kappa - \theta)_+ \sum_{t \in G} |u_t|}.$$

Consequently,

$$\left| \sum_{t \in F} u_t \right|^2 \geq (\kappa - \theta)_+^2 \left(\sum_{t \in G} |u_t| \right)^2 \geq (\kappa - \theta)_+^2 \sum_{t \in G} |u_t|^2.$$

Proof. Project the sum onto ζ :

$$\Re \left(\bar{\zeta} \sum_{t \in F} u_t \right) \geq \kappa \sum_{t \in G} |u_t| - \sum_{t \in B} |u_t|.$$

The first inequality follows. The second uses $(\sum_G |u_t|)^2 \geq \sum_G |u_t|^2$. $\square$

Definition 2.2 (Certified phase mass). For a nonempty family of fibers $\mathcal{G}_X \subseteq \mathcal{N}_X$, choose literal partitions $F_n = G_n \sqcup B_n$ and sector data $(\zeta_n, \kappa_n, \theta_n)$ satisfying [lemma 2.1](#). Put

$$E_X^{\text{cert}} = \sum_{n \in \mathcal{G}_X} \sum_{t \in G_n} |u_t|^2, \quad M_X^{\text{cert}} = \sum_{n \in \mathcal{G}_X} \left(\sum_{t \in G_n} |u_t| \right)^2, \quad \rho_X = \min_{n \in \mathcal{G}_X} (\kappa_n - \theta_n)_+.$$

If $E_X^{\text{cert}} > 0$, define the effective participation

$$R_X^{\text{cert}} = \frac{M_X^{\text{cert}}}{E_X^{\text{cert}}}.$$

It satisfies

$$1 \leq R_X^{\text{cert}} \leq \max_{n \in \mathcal{G}_X} |G_n|.$$

If $E_X^{\text{cert}} = 0$, set $R_X^{\text{cert}} = 0$. For the empty certified family, set $E_X^{\text{cert}} = M_X^{\text{cert}} = \rho_X = R_X^{\text{cert}} = 0$.

Theorem 2.3 (Literal determinant lower certificate). *Every certified family satisfies*

$$\boxed{D_X \geq \rho_X^2 M_X^{\text{cert}} = \rho_X^2 R_X^{\text{cert}} E_X^{\text{cert}}}.$$

The certificate is invariant under a common phase rotation on each fiber and uses the actual post-bin sums.

Proof. Apply the first squared conclusion of [lemma 2.1](#) on each certified fiber and sum. Discarding uncertified fibers is legitimate because their contributions to D_X are nonnegative. The participation bounds follow from

$$\sum_G |u_t|^2 \leq \left(\sum_G |u_t| \right)^2 \leq |G| \sum_G |u_t|^2.$$

$\square$

3 Exponent transfer and compatibility

Recall the high- β scales

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}.$$

TPC-112 requires a lower bound

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2$$

and a zero-mode exponent η_Z satisfying $\lambda_D \leq 2\eta_Z$ [\[2\]](#).

Corollary 3.1 (Three-input exponent ledger). *Fix one predeclared X -indexed selection rule for the certified family, including its sets $G_{X,n}, B_{X,n}$ and sector data. Assume nonnegative exponents $\lambda_E, \lambda_\phi, \gamma_R$ and future literal estimates*

$$E_X^{\text{cert}} \geq X^{-\lambda_E+o(1)} Q_X^3 J_X^2, \quad R_X^{\text{cert}} \geq X^{\gamma_R+o(1)}, \quad \rho_X \geq X^{-\lambda_\phi+o(1)}.$$

Then

$$D_X \geq X^{-(\lambda_E+2\lambda_\phi-\gamma_R)+o(1)} Q_X^3 J_X^2.$$

Thus one admissible nonnegative recorded exponent is

$$\boxed{\lambda_D^{\text{cert}} = (\lambda_E + 2\lambda_\phi - \gamma_R)_+}.$$

This certificate is compatible with a separately proved zero-mode exponent precisely when

$$\boxed{\lambda_D^{\text{cert}} \leq 2\eta_Z}.$$

Proof. Insert the two assumed estimates into [theorem 2.3](#), then apply the TPC-112 cone. $\square$

Remark 3.2 (No double charging). The phase loss occurs twice because the determinant energy is a squared magnitude. The participation gain is used once and may not be recycled as physical synthesis slack. Conversely, equality in the determinant cone is permitted with zero determinant reserve, while the independent physical condition remains strictly $\Lambda_{\text{phys}} < 1/400$.

Remark 3.3 (Noncircular invocation). The three estimates in [corollary 3.1](#) must be proved on the same predeclared certified family and without invoking the target lower bound for D_X . In particular, defining either γ_R or λ_ϕ from the already aggregated value D_X/E_X would merely rename the desired conclusion.

Corollary 3.4 (Participation is indispensable below loss one). *The imported diagonal envelope*

$$E_X^{\text{cert}} \leq E_X \ll X^{-1+o(1)} Q_X^3 J_X^2$$

shows that a certificate with $\lambda_D^{\text{cert}} < 1$ must obtain a positive power from effective participation after paying the phase loss. More precisely, the three-input ledger requires

$$\gamma_R > \lambda_E + 2\lambda_\phi - 1.$$

Proof. The diagonal ceiling forces every matching lower exponent λ_E to be at least 1, up to $o(1)$. Rearrange $\lambda_E + 2\lambda_\phi - \gamma_R < 1$. $\square$

4 Sharp limitations

Proposition 4.1 (Magnitudes do not certify a phase margin). *For every $m \geq 2$ there are m unit-modulus coefficients on one fiber whose sum is zero. Hence active support, equal magnitudes, fiber multiplicity, and diagonal energy alone do not imply $\rho_X > 0$, $C_X > 0$, or $D_X > 0$.*

Proof. Use the m -th roots of unity. $\square$

Proposition 4.2 (A half-plane condition is genuinely additional). *If all coefficients of a fiber lie in an angular sector of width $\Delta < \pi$, then [lemma 2.1](#) applies with $B = \emptyset$ and*

$$\kappa = \cos(\Delta/2) > 0.$$

No sector of width π or more gives a uniform positive lower bound, since opposite phases can cancel.

Proof. Rotate the sector midpoint to the positive real axis. Every coefficient then has real projection at least $\cos(\Delta/2)$ times its modulus. Opposite unit vectors prove the endpoint obstruction. $\square$

The literal phases contain the fixed h_0 , reconstruction multiplier, affine character, and arithmetic signs. Establishing a growing phase margin for them may be as hard as controlling the signed arithmetic directly. The certificate is a target, not a random-phase heuristic.

5 Certificate and claim boundary

The deterministic script `experiments/tpc121_phase_coherence_audit.py` verifies:

1. the mean-fluctuation and weighted-coherence identities;
2. the sector lower bound on a nonsymmetric complex example;
3. exact cancellation by equal-modulus roots of unity; and
4. the participation ratio, a two-fiber minimum-margin test, the diagonal-ceiling requirement, the determinant-zero compatibility check, and the exponent ledger

$$\lambda_D^{\text{cert}} = (\lambda_E + 2\lambda_\phi - \gamma_R)_+.$$

Statement	Level and status
Mean-fluctuation splitting and sector lemma	Exact finite-dimensional theorem (L0).
Attachment to TPC-110 actual determinant fibers	Literal fixed- h_0 interface (L1).
Growing lower bound for E_X^{cert}	Open arithmetic input.
Growing participation gain R_X^{cert}	Open equal-determinant coherence input.
Growing positive phase margin ρ_X	Open coefficient-specific arithmetic input.
Fixed- h_0 polynomial saving	Not proved; no L2 advance.

TPC-121 therefore advances the route by decomposing a sufficient lower certificate for the opaque coherent product $D_X = C_X \bar{m}_X^{\text{coh}}$ into three independently auditable inputs: certified diagonal mass, effective participation growth, and a literal fiber phase margin. It does not show that any of them holds. TPC-122 treats the other side of the compatibility cone: conversion of ordered signed-prefix cancellation into a distinguished zero-mode exponent.

Explicit exclusions. No statement proves H3, a Möbius or Liouville cancellation estimate, a positive prime-pair lower bound, a Hardy–Littlewood asymptotic, a parity breach, or the twin-prime conjecture. Nothing is averaged over or specialized to a substitute for the prescribed h_0 .

References

- [1] Liang Wang. Post-bin determinant energy on the actual support: Exact binning svd, kernel angle, and a natural-scale obstruction, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-110-post-bin-determinant-energy. TPC-110.
- [2] Liang Wang. Determinant-zero compatibility at the high-beta endpoint: Exact exponent cone, reserve accounting, and stop certificates, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-112-determinant-zero-compatibility-audit. TPC-112.